How Does Geometric Bias Shape the Denoising Trajectory in Flow-Based Protein Structure Generation?

Shreyas Ravishankar
University of Cambridge
sr2173@cam.ac.uk

Abstract

Proteina is a flow-based generative model for protein backbone structure where geometry enters only as an additive pair bias B to attention scores. We study how this bias interacts with learned content representations across the generation trajectory using four complementary lenses: pre-softmax signal decomposition, post-softmax contact prediction, causal ablation, and intermediate-layer structure decoding. Across four model scales (60M–400M parameters), we find that (i) B-only attention predicts contacts 5–20$\times$ above random at all scales, (ii) B is dispensable during early generation but essential during late refinement — replicating identically across all four models, and (iii) despite this causal equivalence, the descriptive balance varies over 100$\times$ across scales. Every model concentrates geometric dominance in a "geometric interpreter" layer, but which layer this is depends on architecture: Layer 0 in models without triangle updates, deeper layers in models with them. This dissociation between descriptive signal and causal necessity demonstrates the importance of multi-lens interpretability for understanding geometric inductive biases.

1 Introduction

Generative models for protein backbone structure have advanced rapidly, with flow matching emerging as a leading paradigm [8]. Proteina [5] scales flow-based protein generation to state-of-the-art quality by combining a transformer backbone with *Pair Bias Attention* — a mechanism that fuses learned content scores with geometric pair biases.

Unlike SE(3)-equivariant networks [4, 11] or invariant point attention [6], Proteina's transformer layers are standard (non-equivariant) attention and MLP blocks. Geometry enters only through an additive pair bias B derived from pairwise distances and sequence separation — a *soft invariant prior* that encodes geometric information without constraining the network's outputs to be equivariant. This reflects a broader trend away from hard-coded architectural symmetries toward soft geometric priors, as seen in ViT [3], TokenGT [7], and AlphaFold3 [1].

This raises a natural question: **how much work does the geometric bias actually do — and when?** During generation, Proteina samples via SDE integration from pure noise (t=0) to a structured backbone (t=1). At each step, every attention layer computes:

$$A = \mathrm{softmax}(C + B) \tag{1}$$

where $C = QK^\top/\sqrt{d}$ is the content score and B is the pair bias. Since softmax is nonlinear, there is no single correct way to attribute the output to B vs. C. We therefore ask four complementary questions:

1. **Descriptive:** How large is B relative to C in the pre-softmax scores, and how does this evolve?
2. **Functional:** Which component produces more structurally useful attention patterns?
3. **Causal:** When and where is B necessary for generating valid structures?
4. **Representational:** Where in the network does 3D structure emerge?

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We apply all four lenses to four model variants spanning 60M to 400M parameters (Table 1). Our main findings are:

- B-only attention predicts contacts 5–20× above the random baseline at all scales, while C-only is only 2–3× above random (Section 3.2).
- B is dispensable during early generation ($t < 0.5$) but essential during late refinement — and this temporal asymmetry replicates identically across all four models despite >100× differences in descriptive signal (Section 3.3).
- Triangle multiplicative updates do not change *when* B matters but invert *where*: they shift B's critical contribution from early to deep layers (Section 3.3).
- Every model has a "geometric interpreter" layer where $R_c \gg 1$, but which layer this is depends on architecture, not scale (Section 3.1).

2 Methods

2.1 Models and Generation

We analyse four Proteina variants (Table 1). The no-tri models use only pair bias attention; the tri models additionally use triangle multiplicative updates that iteratively refine the pair representation. All models include 10 register tokens [2]. Proteins of length n=100 are generated unconditionally via 100-step SDE sampling (Euler-Maruyama with dt=0.01, noise scale 0.45). The pre-softmax components C and B are captured every 5th step, yielding 20 temporal snapshots. Descriptive and functional experiments use 5 seeds; causal ablations use 3 seeds.

Table 1: Model variants analysed.

Model	Version	Layers	Heads	d_{token}	d_{pair}	Triangle
60M no-tri	v1.3	12	12	512	256	No
200M no-tri	v1.2	15	12	768	256	No
200M tri	v1.1	15	12	768	512	Yes
400M tri	v1.4	18	16	1024	512	Yes

2.2 Metrics

Table 2 defines all metrics with their null baselines.

2.3 Four Analytical Lenses

Lens 1: Descriptive. We measure $R_c = \|\tilde{B}\|_F / \|\tilde{C}\|_F$, the row-centred Frobenius norm ratio (Table 2). Row-centering removes the softmax-invariant component: since $\mathrm{softmax}(\mathbf{x} + c) = \mathrm{softmax}(\mathbf{x})$, row-constant energy inflates the raw ratio R without affecting attention. Figure 3 justifies this choice. We also decompose R_c by sequence separation ($|i-j|$) to test whether B's role differs for local vs. long-range interactions.

Lens 2: Functional. Following Rao et al. [10], we evaluate Contact Precision@$L/5$ for $\mathrm{softmax}(C + B)$, $\mathrm{softmax}(B)$, and $\mathrm{softmax}(C)$. Ground truth contacts are defined retrospectively from each model's own generated structure. The random baseline (dashed line in figures) is the contact density among eligible pairs.

Lens 3: Causal. We zero out B under six conditions: full ablation, early-only B ($t \le 0.5$), late-only B ($t > 0.5$), Layer-0 ablation, deep-layer ablation (second half of layers), and baseline. We additionally ablate each model's peak-R_c layer individually to test whether descriptive dominance implies causal importance.

Lens 4: Representational (structure lens). Inspired by the logit lens [9], we apply the coordinate decoder to intermediate layer representations, producing a per-layer 3D prediction that we compare to the final output via RMSD and contact similarity.

Table 2: Summary of metrics. "Baseline" is the expected value under a null model.

Metric	Definition	Range	Baseline
Descriptive			
R_c (row-centred logit dominance)	$\|\tilde{B}\|_F/\|\tilde{C}\|_F$ where $\tilde{X}$ subtracts each row's mean. Accounts for softmax invariance to row-constant shifts.	$[0, \infty)$	$R_c = 1$ (balanced)
Functional			
Precision@$L/5$	Fraction of the top-k attention-ranked residue pairs ($k = n/5$) that are true contacts (C$\alpha < 8\,\text{Å}$, $\lvert i-j\rvert \geq 6$). Symmetrised and APC-corrected following Rao et al. [10]. Short-range pairs excluded (trivially close along backbone).	$[0, 1]$	Contact density (∼1–3%)
Causal			
R_g (radius of gyration)	$\sqrt{\frac{1}{n}\sum_i \|\mathbf{r}_i - \bar{\mathbf{r}}\|^2}$	$(0, \infty)$ nm	∼1.3–1.4 nm
Steric clashes	Cα pairs with distance $< 2\,\text{Å}$, $\lvert i-j\rvert \geq 3$	$[0, \infty)$	0
Representational (structure lens)			
RMSD to final	Kabsch-aligned RMSD of intermediate-layer decoded coordinates to the final-layer prediction	$[0, \infty)$ nm	0 at final layer
Contact similarity	Jaccard index of contact maps between intermediate and final layer predictions	$[0, 1]$	1 at final layer

3 Results

3.1 Descriptive: Every Model Has a Geometric Interpreter

Figures 1 and 2 show R_c across the generation trajectory.

Every model concentrates geometry dominance in one or a few layers (Figure 2). In the 60M no-tri model, Layer 0 dominates ($R_c \approx 35$ at t=1); in the 200M no-tri model, Layers 0 and 2 are elevated but weaker ($R_c \approx 4$); in the 400M tri model, Layer 2 dominates ($R_c \approx 29$). The 200M tri model distributes geometry across many layers (L4, L7, L9, L13), with L9 peaking at $R_c \approx 27$. The remaining layers in all models stay near $R_c \approx 1$ (balanced).

Which layer is the geometric interpreter depends on architecture, not just scale. Both no-tri models concentrate in early layers (L0–L2); both tri models shift the peak to deeper layers, consistent with triangle updates amplifying B's role where they operate. The 200M tri model is distinctive in distributing geometry across many layers rather than concentrating it.

Why R_c rather than R? Figure 3 compares raw R and row-centred R_c for the 60M model. At t=1, $R \approx 5.7$ but $R_c \approx 35$: the content score develops large row-constant components that inflate its Frobenius norm without affecting attention. Under the more principled R_c, geometry dominance in Layer 0 is 6× greater than the raw metric suggests.

Sequence separation modulates the balance. Figure 4 decomposes R_c by residue separation in the peak layer. The 60M model shows the strongest separation dependence; the 200M no-tri and 400M tri models show the same qualitative pattern (long-range > local early, reversing late) but weaker. The 200M tri model again stands out as the exception, with a distinctive non-monotonic trajectory.

3.2 Functional: B Predicts Contacts Far Better Than C

Figure 5 evaluates contact prediction across all four models. At t=1, bias-only precision is 5–20× above the random baseline at all scales: 0.26 for the 60M and 200M models, 0.46 for the 400M tri

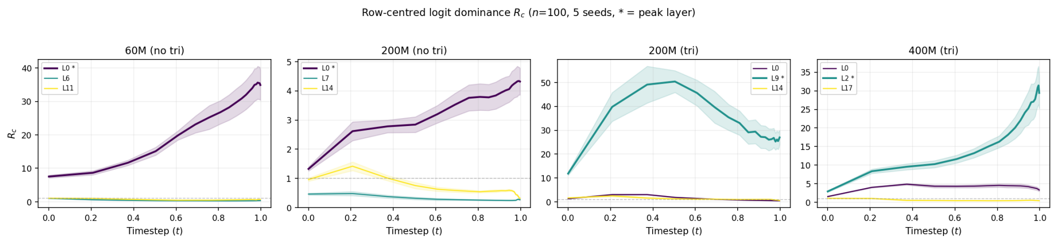

Figure 1: Row-centred logit dominance R_c for key layers in each model (n=100, 5 seeds). Layers selected per model: peak R_c layer (marked with $*$) plus first/last for contrast. Dashed line: $R_c = 1$ (balanced).

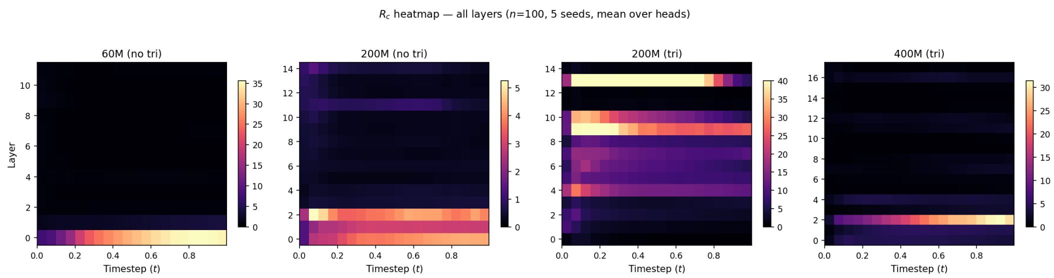

Figure 2: R_c across all layers and timesteps (mean over heads). Each model concentrates geometry dominance in different layers: L0 in 60M, L0/L2 in 200M no-tri, distributed in 200M tri, L2 in 400M tri.

model (random $\approx$ 0.01–0.03). Content-only precision is 2–3$\times$ above random ($\sim$0.05–0.10) — above chance but far weaker than B.

This functional gap is **scale-invariant**: the 200M no-tri model has $R_c \approx 1.3$ in Layer 0 (near-balanced) yet B-only contact prediction is just as superior as in the 60M model where $R_c \approx 7.5$. Descriptive signal magnitude does not predict functional utility.

Full attention slightly underperforms bias-only, suggesting C occasionally opposes B by encoding complementary non-contact relationships.

3.3 Causal: B Matters Only at the End

Table 3 shows structural quality under ablation. Full removal of B is catastrophic in every model (thousands of clashes).

The central causal finding is a **sharp temporal asymmetry that replicates across all four models**: giving B only during early generation ($t \leq 0.5$) then removing it is indistinguishable from never having B at all. Conversely, withholding B during early generation then activating it ($t > 0.5$) nearly preserves structural quality. This implies a single temporal boundary around $t \approx 0.5$: B is dispensable before it and essential after.

Triangle updates invert the layer-specific profile without changing the temporal pattern. In no-tri models, Layer-0 ablation is the most damaging single-layer intervention ($R_g = 0.81$ nm for 60M). In tri models, Layer-0 ablation is nearly harmless ($R_g \geq 1.1$ nm) while deep-layer ablation is catastrophic ($R_g \leq 0.12$ nm, thousands of clashes). The 400M tri replicates the 200M tri pattern, confirming this is driven by triangle updates, not scale.

Descriptive dominance does not imply causal necessity for individual layers. Ablating each model's peak-R_c layer (Table 4) produces only moderate damage in all cases: R_g drops by 5–18% but structures remain valid (0–2 clashes). Even Layer 2 in the 400M tri model, which has $R_c = 29.4$ (comparable to Layer 0 in the 60M model), is individually dispensable. By contrast, ablating the *collective* deep layers is catastrophic in tri models. The geometric bias is distributed enough that

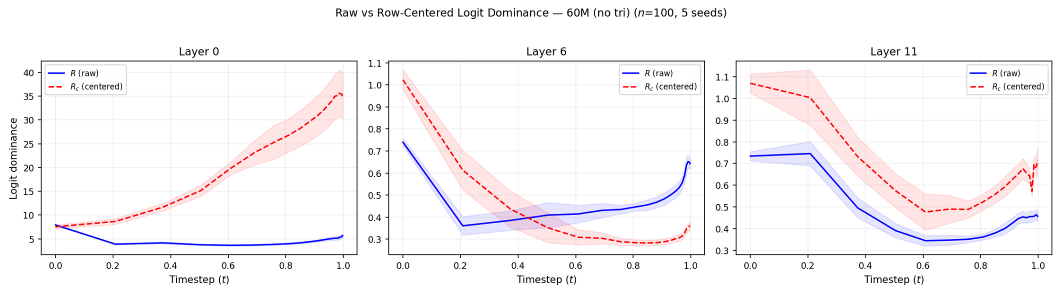

Figure 3: Raw R (blue) vs. row-centred R_c (red) for the 60M model. In Layer 0, R_c diverges from R at late timesteps, revealing that C develops softmax-invariant row-constant energy.

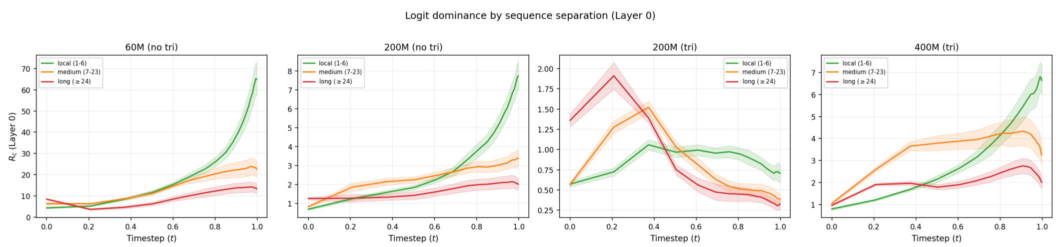

Figure 4: R_c by sequence separation in the peak layer for all four models (5 seeds).

no single high-R_c layer is essential on its own — it is the aggregate contribution across layers that matters.

3.4 Representational: Structure Emerges Only at the Output

Figure 6 applies each model's coordinate decoder to intermediate layer representations. Middle layers produce representations *furthest* from the final output (highest RMSD), while early-layer representations are closer — the network warps representations through an internal space not aligned with coordinate space. Contact similarity to the final structure is near zero until the last 1–2 layers, concentrating structural specificity at the output. The radius of gyration panel shows that intermediate layers decode to structures with varying compactness, with early layers at early timesteps producing unexpectedly compact structures. Triangle models show a more gradual RMSD gradient, consistent with the pair representation providing an additional channel for structural information.

4 Discussion

The geometric bias is essential late in generation, regardless of scale or architecture. The temporal ablation pattern — early-only $B \equiv$ full ablation; late-only $B \approx$ baseline — replicates identically across all four models despite $>100\times$ differences in the descriptive metric R_c. This is our strongest finding: descriptive signal magnitude is decoupled from causal necessity. The content score C does not learn to replicate B's structural signal at any scale — C-only attention remains poor at contact prediction even in models where C's norm exceeds B's.

Triangle updates determine where B is critical, not when. Without triangle updates, Layer-0 ablation is most damaging and deep ablation is moderate. With triangle updates, this inverts: Layer-0 ablation is harmless while deep ablation is catastrophic. Triangle updates create a deep-layer information bottleneck through the pair representation, concentrating B's causal contribution in those layers. The temporal pattern is unaffected.

No single layer is indispensable. Despite concentrating geometric signal ($R_c \gg 1$), the peak-R_c layer can be individually ablated with only moderate damage (Table 4). Catastrophic failure requires ablating *many* layers collectively. The geometric bias functions as a distributed system: each layer's contribution is replaceable, but the aggregate is essential.

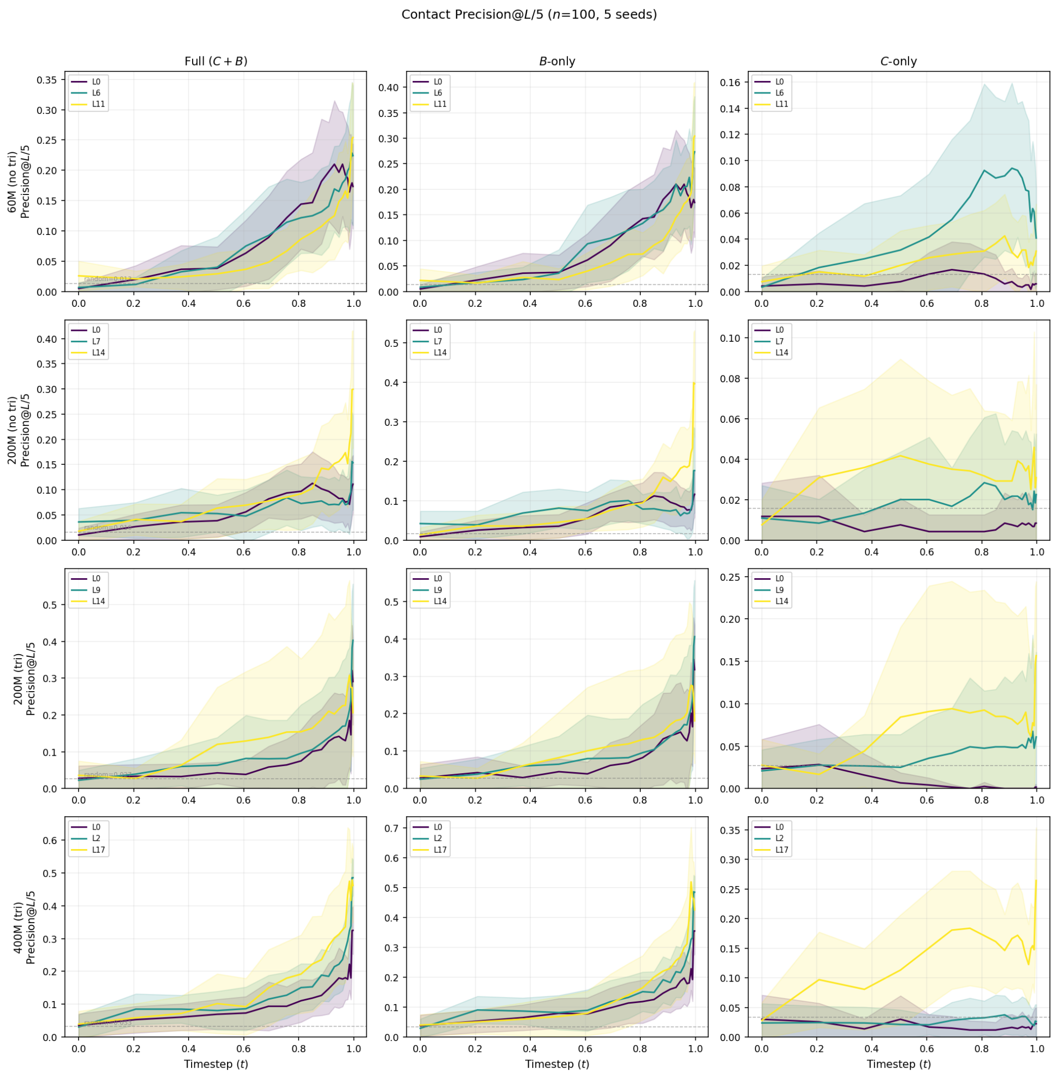

Figure 5: Contact Precision@$L/5$ (n=100, 5 seeds). Dashed line: random baseline (contact density). B-only is 5–20× above random at all scales.

Implications for architecture design. Soft geometric priors (additive biases) are effective alternatives to hard constraints (equivariant layers), provided they are present during late refinement. Conclusions about geometric bias importance drawn from magnitude-based metrics alone may not generalise across model scales — multi-lens analysis is necessary.

Limitations and future work. All analyses use unconditional generation with retrospective ground truth; extending to conditional generation would test whether these findings hold when external structural ground truth is available. The Frobenius norm ratio does not capture the cross-term interaction $\langle \tilde{B}, \tilde{C} \rangle$; a full variance decomposition could clarify whether B and C reinforce or oppose each other.

5 Conclusion

The geometric pair bias in Proteina is dispensable during early generation but essential during late structural refinement — a causal finding that replicates across four model scales despite $>$100× variation in the descriptive geometry–content balance. Triangle updates do not change when B matters but invert where, shifting the critical contribution from early to deep layers. No single

Table 3: Structural quality under bias ablation (n=100, 3 seeds). Temporal asymmetry replicates across all four models. Bold: triangle update effect on layer-specific profile.

	60M no-tri		**200M no-tri**		**200M tri**		**400M tri**	
Condition	R_g	Clash	R_g	Clash	R_g	Clash	R_g	Clash
Baseline	1.44	0	1.36	0	1.42	0	1.27	1
Full ablation	0.12	3532	0.14	2968	0.07	4702	0.11	4021
Early-only B	0.12	3532	0.14	2967	0.07	4702	0.11	4021
Late-only B	1.19	0	1.19	4	1.12	2	1.17	1
Layer-0 abl.	0.81	1	0.91	5	**1.24**	**1**	**1.13**	**0**
Deep abl.	0.60	23	0.63	32	**0.05**	**4752**	**0.12**	**3560**

Table 4: Ablating each model's peak-R_c layer (3 seeds). No single geometric interpreter layer is catastrophically important on its own, despite high R_c.

Model	Peak layer	Peak R_c (t=1)	R_g (ablated)	R_g (baseline)	Clashes
60M no-tri	L0	34.9	0.81	1.44	1
200M no-tri	L2	3.9	1.28	1.36	1
200M tri	L9	27.0	1.17	1.42	1
400M tri	L2	29.4	1.13	1.27	1

"geometric interpreter" layer is individually indispensable, but the aggregate geometric signal across layers is essential. Soft geometric priors play functionally irreplaceable roles in flow-based protein generation, and multi-lens interpretability is necessary to reveal this.

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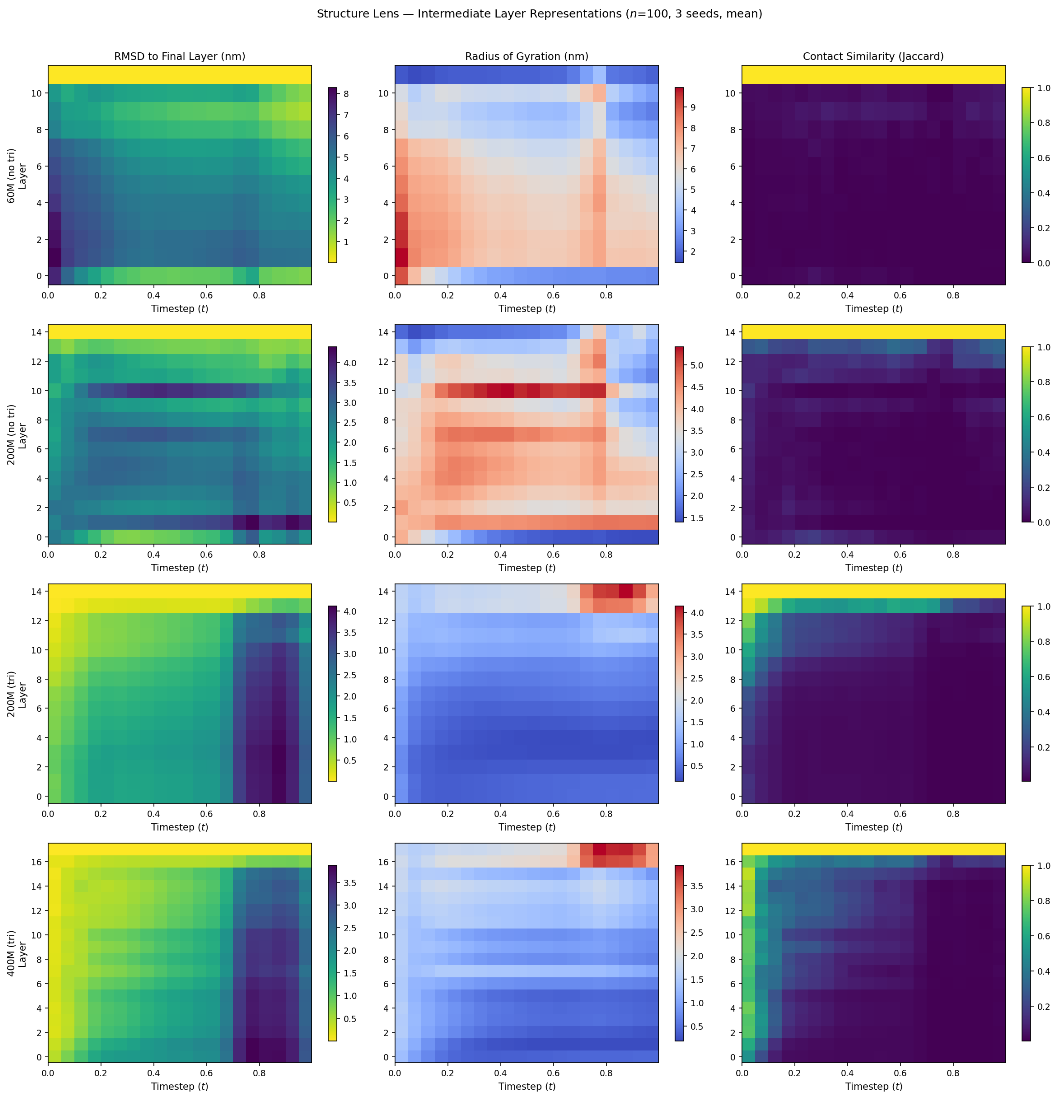

Figure 6: Structure lens (3 seeds, mean). **Left:** RMSD to final-layer prediction — middle layers are furthest. **Centre:** Radius of gyration of decoded structures. **Right:** Contact similarity (Jaccard) — near zero until the last layers.